Economic Rankings Need an Operational Contract

Price, Effort, and Reliability on DeepSWE v1.1

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Abstract

DeepSWE ranks coding-agent capability, but its trial artifacts also permit economic comparison. We analyze 18,396 scored attempts from 41 model-effort configurations on 113 tasks. Realized cost per successful completion (CPSC) equals mean attempt cost divided by pass rate, so its point estimate and rank are recoverable from leaderboard aggregates. Trial records matter for paired suite-composition uncertainty, equal-task weighting, recorded-resource intensity, and policy reselection. Three results follow. First, within GPT-5.6 Sol, high and xhigh effort are cheaper and use fewer recorded resources than max in every task- and repository-bootstrap replicate, while paired solve-rate intervals include zero. Comparing Luna max with Sol high leaves the invoice difference unresolved and strongly favors Sol on recorded work and elapsed time. Second, reported-dollar and recorded-work frontiers disagree. Luna max costs 0.390 times Sol max per success while using 1.80 times the agent steps and 2.12 times the summed reported token counters. Third, no configuration is preferred independently of policy: reliability floors and proxy failure charges change the selected configuration, and high floors frequently leave the feasible set unstable or empty. DeepSWE therefore yields several well-defined economic rankings but does not identify an operational recommendation without a declared workload, reliability requirement, failure valuation, and price basis.

1 Introduction

DeepSWE v1.1 asks which fixed coding-agent configuration solves the largest share of 113 original, functionally verified software-engineering tasks [5]. Its public artifact also contains trial spend, steps, tokens, and repeated attempts. This permits a different, deliberately open question: *what happens when DeepSWE is analyzed as stochastic spend and work per verified completion?*

For configuration m, realized cost per successful completion is

$$\widehat{\mathrm{CPSC}}_m^{\mathrm{real}} = \frac{\text{total scored spend}_m}{\text{verified successes}_m} = \frac{\text{mean attempt cost}_m}{\widehat{p}_m}. \tag{1}$$

The identity is important. The point estimate and its rank are recoverable from DeepSWE's published mean attempt cost and pass rate. The contribution cannot be the division itself. Trial records are useful because they support task-paired uncertainty, task-balanced workloads, outcome-conditioned spend, solved-task overlap, resource consumption, exclusion audits, and policy reselection.

The cleanest economic contrast is within one model and provider. Sol high and xhigh reduce invoice, recorded work, and elapsed time relative to max, while their point solve-rate deficits remain unresolved under suite resampling. A cross-family example then shows why the surfaces differ: Luna

Table 1: Contribution map. Aggregate fields identify several point rankings; trial records identify their suite-composition stability and additional economic surfaces. Neither identifies a deployment contract.

Recoverable from leaderboard aggregates	Requires trial or task records
CPSC point estimates and ranks	Task- and repository-cluster paired intervals
Unconstrained minimum CPSC	Reliability-floor reselection frequencies
Pass–CPSC frontier	Recorded-resource intensities and frontiers
	Equal-task estimands and eligibility changes
	Matched solved-task and exclusion diagnostics

max is much cheaper than Sol max in reported dollars but consumes materially more agent steps and reported token counters. Comparing Luna max with Sol high nearly eliminates the invoice gap while Sol uses less work and time. Effort setting and model family therefore affect different economic surfaces. The primary economic contrasts are within the OpenAI GPT-5.6 generation; cross-provider dollars remain descriptive invoice surfaces.

The economic answer is also policy-conditional. An unconstrained minimum rewards configurations with solve rates as low as 35%. Raising the reliability requirement changes the selected configuration and eventually leaves no eligible option. Retrospective proxy charges for failure move the ranking again. The correct conclusion is not that DeepSWE names a new universal winner. It is that the benchmark contains multiple economic surfaces, and that a recommendation is not identified until an operational contract specifies the workload, reliability requirement, failure consequence, and price basis.

2 Related Work and Claim Boundary

Software-engineering benchmarks. SWE-bench established repository-level issue resolution against executable tests as an evaluation problem [6]. DeepSWE authors tasks from scratch across 91 repositories and five languages and grades them with purpose-built functional verifiers [5]. SWE-Lancer connects freelance software tasks to historical payouts, measuring task-side economic value [7]. We instead analyze model-side benchmark spend. We do not assume that a completion has the same value across benchmarks.

Cost-aware inference. FrugalGPT and RouteLLM study cost-quality tradeoffs through routing [2, 8]. Cost-of-Pass formalizes expected monetary cost per correct solution and an economic frontier [4]. Under a fixed attempt distribution, realized CPSC is the same cost-over-success object. Coding-agent work also studies token-consumption variance and model-level token efficiency [1]. We do not claim novelty for cost divided by success probability. The contribution here is empirical and diagnostic: transport the accounting to an independent frontier benchmark, separate invoice from resource appetite, and identify which economic claims remain policy-dependent.

Claim boundary. The analysis can show that the accounting and task- and repository-clustered stability procedures behave coherently on DeepSWE. It cannot validate a production reliability requirement, calibrated task-specific failure values, a stateful repair policy, a common-price ranking, or production return on investment. Exact reconciliation and tests support implementation correctness; DeepSWE supplies a transport study, not construct or decision validation for another benchmark.

3 Data, Estimands, and Methods

3.1 Cohort, cost, and exclusions

The analysis uses content-hashed DeepSWE v1.1 artifacts frozen on July 14, 2026 [3]. The trial file contains 18,522 rows. Requiring the DeepSWE source, full evaluation scope, and score-inclusion flag leaves 18,396 scored attempts across 41 configurations. Model-caused timeouts and context-window failures remain scored. We exclude 126 rows marked as infrastructure failures: 73 routing 404s, 36 provider timeouts, 11 verifier timeouts, and six other errors. Exclusions are provider-skewed and are audited separately.

Twenty-one scored rows lack `cost_usd`: 20 failed Fable rows on one task and one successful Sonnet row. The primary analysis retains their outcomes and imputes the configuration mean cost. Frozen sensitivities charge zero, use the configuration-outcome median, or remove the row from both numerator and denominator. The primary pipeline reproduces all official pass rates, mean costs, and implied CPSC values within 3.6×10^{-15}.

3.2 Observed-attempt and equal-task estimands

Let $Y_{mti} \in \{0, 1\}$ be verified success and $C_{mti} \geq 0$ reported spend for configuration m, task t, and scored attempt i. With S_m successes and F_m failures,

$$\widehat{\mathrm{CPSC}}_m^{\mathrm{real}} = \frac{\sum_{t,i} C_{mti}}{S_m} = \widehat{\mu}_m^S + \frac{F_m}{S_m}\widehat{\mu}_m^F, \tag{2}$$

where $\widehat{\mu}_m^S$ and $\widehat{\mu}_m^F$ are mean spend conditional on success and failure. The second term is the realized reliability tax.

Equation 2 weights tasks by their number of surviving scored attempts. To represent a declared equal-task workload, we also compute

$$\widehat{\Theta}_m = \frac{\sum_t w_t \,\overline{C}_{mt}}{\sum_t w_t \,\overline{Y}_{mt}}, \qquad w_t = 1/113, \tag{3}$$

where each bar averages valid attempts within a task. The estimands coincide when every configuration-task cell has the same attempt count. Here 4,520 of 4,633 cells have four attempts; 111 have one to three; and two Opus max cells are absent. We report the 113-task estimand for the 40 complete configurations, a common 111-task basket for all 41, and the observed-attempt invoice ratio.

Realized CPSC does not identify a sequential retry policy. Attempts on a task are dependent, and the artifact does not declare a randomized retry order or stopping rule. We also report empirical task coverage, the share of tasks solved at least once in the observed attempts, without calling it production pass@k.

3.3 Recorded work and price basis

For each recorded resource $R \in \{$agent steps, input, cache, output tokens, agent seconds, trial seconds$\}$, resource intensity is

$$\widehat{I}_m^R = \frac{\sum_{t,i} R_{mti}}{S_m}. \tag{4}$$

We additionally sum the three reported token counters per success. These are behavioral, billing, and elapsed-time surfaces, not harmonized compute. Tokenizers, cache semantics, reasoning-token exposure, and service conditions may differ by model and provider, so cross-family levels are descriptive. Of 18,396 scored rows, 18,395 contain all four step/token fields; one successful Sonnet 5 high row lacks all four. We therefore mark that configuration's corresponding totals undefined and exclude it from those ranks, pairwise ratios, and frontiers. Agent and trial duration are complete for all scored rows. Reported dollars answer an invoice question under the artifact's provider prices. They do not reveal provider cost-to-serve or margin.

3.4 Task-mix diagnostics and uncertainty

For every pair we compute solved-task overlap and recompute CPSC on tasks both configurations solve at least once. This matched set conditions on outcomes and is a composition diagnostic, not a deployment estimand. We also form panel-solvedness strata from the number of configurations that solve each task. To examine GPT-5.6 Luna, Sol, and Terra without circularly using their outcomes to define their own strata, the primary mechanism diagnostic excludes all 15 GPT-5.6 configurations when assigning 20 rare, 30 contested, and 63 common tasks. The labels remain retrospective panel-solvedness categories, not intrinsic difficulty or a preregistered pressure scale.

The highlighted Sol-effort and Luna–Sol case studies were selected after inspecting the panel and are interpreted as exploratory mechanism contrasts. Their intervals measure suite-composition stability, not confirmatory error control across all possible pairwise comparisons.

Primary uncertainty uses 10,000 task-cluster bootstrap replicates with seed 20260714. Each replicate samples 113 task IDs with replacement and retains every scored attempt for each selected task. Paired contrasts use the same task sample, and reported 95% intervals are the 2.5th and 97.5th percentiles. The task-balanced sensitivity recomputes task-level means, equal-task eligibility, and policy selection after the same resampling step. Intervals therefore measure sensitivity to suite composition, not all future tasks or rollout randomness. Reliability policies and economic winners are reselected inside each replicate.

Because 113 tasks come from 91 repositories, a reviewer-requested sensitivity also samples 91 repository IDs with replacement and retains every task and attempt within each selected repository. Seventy-five repositories contribute one task, 12 contribute two, three contribute three, and one contributes five. The sensitivity uses 10,000 replicates with seed 20260715 and recomputes the four highlighted contrasts plus policy selection at the 70% and 75% floors. With 113 task clusters and only four dependent attempts within most configuration–task cells, several-point solve-rate differences remain sensitive to suite composition even when cost ratios are highly stable.

3.5 Reliability and failure-price policies

For required solve rate r, the point policy is

$$m^*(r) = \arg \min_{m:\widehat{p}_m \geq r} \widehat{\mathrm{CPSC}}_m^{\mathrm{real}}. \tag{5}$$

We evaluate $r = 0, 0.05, \ldots, 0.75$. We report selection frequency over all replicates and conditional on at least one eligible configuration. These are task-mix robustness frequencies, not posterior model probabilities.

A calibrated task-specific failure charge B_t is unavailable. We therefore run two explicitly retrospective proxy sensitivities. The full-basket version uses median observed Sol max spend by task. A stricter 97-task version uses median successful Sol max spend only on tasks Sol solves. Failed attempts are charged 0.5, 1, or 2 times the proxy. These outputs are counterfactual proxy charges, never reference-budget estimates.

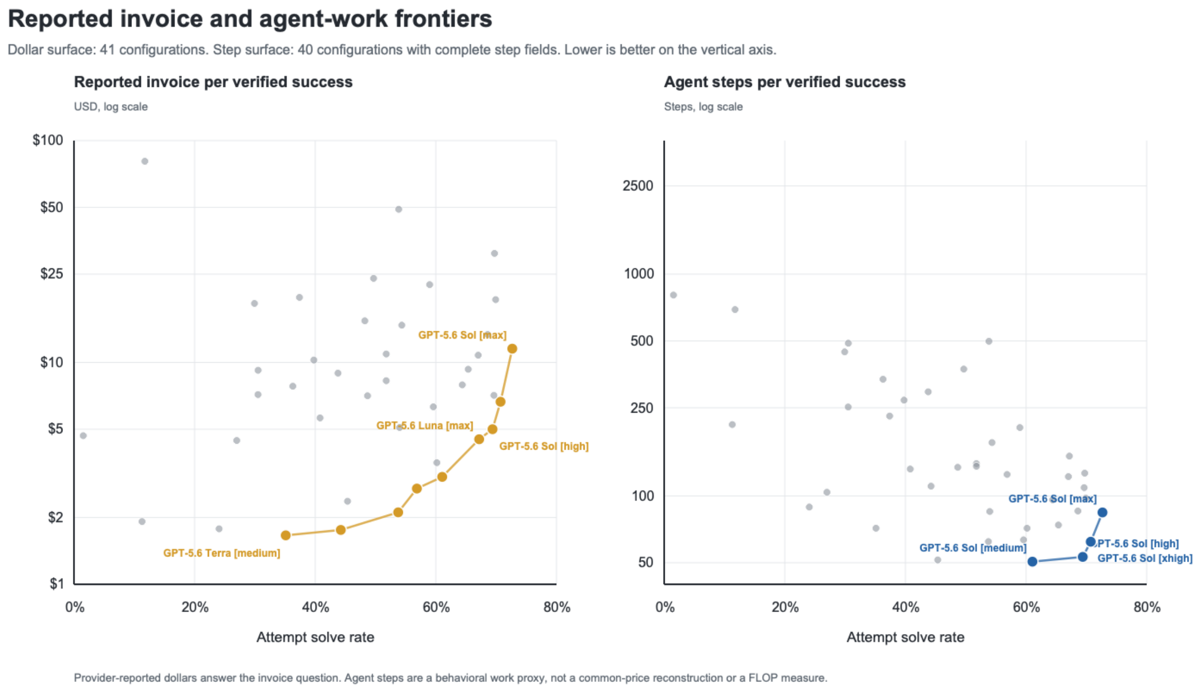

Figure 1: Pass rate versus reported-dollar CPSC (left) and agent steps per success (right). The dollar panel includes all 41 configurations; the step panel includes the 40 configurations with complete step fields. Colored points are nondominated within each surface. The frontier changes when dollars are replaced by recorded work. Dollar comparisons use reported provider prices; steps are not harmonized compute.

4 Results

4.1 The full panel separates invoice from work

Sol max leads attempt solve rate at 72.7%; the highest-pass family panel is reported descriptively in table 6 in the appendix. Across the full 41-configuration menu, unconstrained Terra medium minimizes CPSC at $1.66 with only 35.1% pass rate. The all-configuration Pareto surfaces are more revealing than either row. Nine configurations are nondominated on the pass–CPSC frontier, spanning GPT-5.6 Luna, Terra, and Sol. In contrast, the four step-per-success frontier configurations are all Sol; every input-, output-, or cache-counter frontier point is also Sol. The descriptive summed-counter frontier agrees. Figure 1 visualizes the pass–CPSC and pass–step-intensity surfaces. A configuration can be invoice-efficient without minimizing recorded agent work.

This finding is not a claim that Sol is intrinsically cheaper to serve. It is a decomposition of what the artifact can identify. Dollar efficiency equals observed resource appetite filtered through provider-specific units and prices. The data measure both sides imperfectly, but they are sufficient to reject the idea that one universal notion of “efficiency” explains the ranking.

4.2 Sol max’s cost premium is resolved; its solve-rate gain is not

Within one model and provider, Sol high, xhigh, and max offer the cleanest effort comparison (table 2). Max has the highest point solve rate. However, paired task-bootstrap intervals for high-minus-max ($[-7.7, +1.1]$ points) and xhigh-minus-max ($[-5.6, +1.7]$) include zero. Meanwhile high and xhigh are cheaper than max in every replicate and use fewer steps, individual token counters, and accumulated seconds with all paired ratio intervals below one.

Table 2: Within-Sol effort comparison. The counter column is a descriptive sum of reported input, cache, and output counters; seconds are accumulated trial duration per verified success.

Configuration	Pass	CPSC	Steps/success	Sum counters/success	Trial s/success
Sol high	69.4%	$5.00	53	7.5M	1,004
Sol xhigh	70.7%	$6.65	62	11.7M	1,273
Sol max	72.7%	$11.54	84	21.2M	1,693

Repository clustering preserves the result. High-minus-max has solve-rate interval $[-7.8, +1.3]$ points and CPSC-ratio interval $[0.399, 0.470]$; xhigh-minus-max has $[-5.6, +1.6]$ points and $[0.537, 0.619]$. Both alternatives are cheaper and leaner than max in all 10,000 repository-bootstrap replicates. The interpretation is asymmetric: max's additional dollar, recorded-resource, and time cost is resolved; its solve-rate advantage is not. A larger evaluation could resolve the capability difference. For this suite under pure CPSC, maximum effort appears overprovisioned unless a decision maker values the unresolved point advantage enough to pay the premium.

4.3 Luna–Sol is a price-versus-work reversal

Table 3 compares Luna max with capability-leading Sol max. The point solve-rate gap is −5.5 percentage points, with paired task-bootstrap interval $[-12.1, +1.1]$; this is unresolved, not evidence of equivalence. Luna's CPSC ratio is 0.390 [0.345, 0.439], and Luna is cheaper in all 10,000 task-mix replicates. Yet Luna consumes materially more of every displayed step/token counter per success. Its trial-seconds ratio, 1.08 [0.97, 1.20], is unresolved. The result is an invoice advantage, not a lower-recorded-work advantage.

Table 3: Luna max versus Sol max. Ratio intervals are paired 95% task-cluster bootstrap intervals. A ratio below one favors Luna.

Metric	Luna max	Sol max	Luna/Sol contrast
Attempt solve rate	67.2%	72.7%	−5.5 pp $[-12.1, +1.1]$
Any-success task coverage	102/113	97/113	descriptive +5 tasks
Reported CPSC	$4.51	$11.54	0.390 [0.345, 0.439]
Agent steps/success	151	84	1.80 [1.62, 1.99]
Input counter/success	23.0M	10.9M	2.11 [1.87, 2.38]
Output counter/success	0.109M	0.083M	1.32 [1.20, 1.46]
Sum of counters/success	44.9M	21.2M	2.12 [1.88, 2.39]
Trial seconds/success	1,830	1,693	1.08 [0.97, 1.20]

The repository bootstrap gives the same interpretation: Luna's solve-rate gap is $[-11.4, +0.6]$ points, its CPSC ratio is $[0.350, 0.434]$, and its step and summed-counter ratios remain strictly above one. The task-cluster result is not an artifact of treating related tasks as independent clusters.

The solved-task composition does not erase the reversal. Luna and Sol solve 92 tasks in common (Jaccard overlap 0.86). On that post-outcome intersection, Luna's CPSC is $3.88 versus Sol's $9.72, while Sol's solve rate is higher, 85.8% versus 77.2%. This is a diagnostic, not a policy for unseen tasks.

The noncircular GPT-5.6 group-out strata also do not support a simple "Luna solves only the easy subset" story. On the 20 tasks deemed rare by the other 26 configurations, Luna succeeds on 39 of 80 attempts and reaches 16 tasks; Sol succeeds on 34 and reaches 11. Sol leads attempt solve rate on the contested and common strata, while Luna remains cheaper in all three. This retrospective coverage difference is not a resolved behavioral mechanism.

What explains the dollar reversal? The data answer only part of the question. Luna does more recorded work, so lower resource consumption is not the mechanism. The residual necessarily lies in how those resources map to reported dollars, including model-specific unit prices, cache rules, and unobserved token categories. Provider cost-to-serve, parameter count, utilization, and margin are absent. Those are plausible explanations, not findings.

The effort-tuned cross-model comparison changes the headline gap. Luna max versus Sol high has a CPSC ratio of 0.901 [0.798, 1.018], which crosses parity, and a solve-rate difference of -2.2 points $[-8.6, +4.0]$. The repository-cluster CPSC interval is [0.807, 1.010], also crossing parity. Sol high uses 0.35 times the steps [0.32, 0.39] and 0.55 times the trial seconds [0.49, 0.61], with lower individual input and output counters; the descriptive summed-counter ratio is 0.17 [0.15, 0.19]. Thus the Luna–Sol-max invoice gap is partly an effort-setting result. It largely disappears when Sol is evaluated at high rather than max effort, while the work and elapsed-time gap becomes larger.

4.4 Observed failure spend is not an independent signal

Across configurations, mean failed-attempt spend is 0.72 to 1.23 times mean successful-attempt spend, with median 1.02. The realized reliability-tax share therefore correlates 0.995 with failure rate. Under DeepSWE's fixed attempt policy, failures usually consume about as much reported spend as successes. The decomposition is correct and useful for auditing, but its share is mostly a rescaling of unreliability rather than an independent behavioral signal. An exogenous failure consequence could behave differently; DeepSWE does not identify one.

5 Why the Economic Winner Is Policy-Conditional

5.1 Reliability changes the feasible set

The unconstrained point minimum, Terra medium, passes 35.1% of attempts. As the floor in equation (5) increases, the point winner moves through Terra, Luna, and Sol configurations. At 65%, Luna max minimizes observed CPSC; at 70%, Sol xhigh does; no point estimate reaches 75%.

Bootstrap reselection shows that high-reliability decisions are unstable (table 4). Sol high can be selected at 70% despite its 69.4% full-sample pass rate because pass rate and eligibility are recomputed on each resampled task mixture. Selection shares combine uncertain eligibility near a hard threshold with uncertain CPSC ordering. They are not posterior probabilities that a model is best.

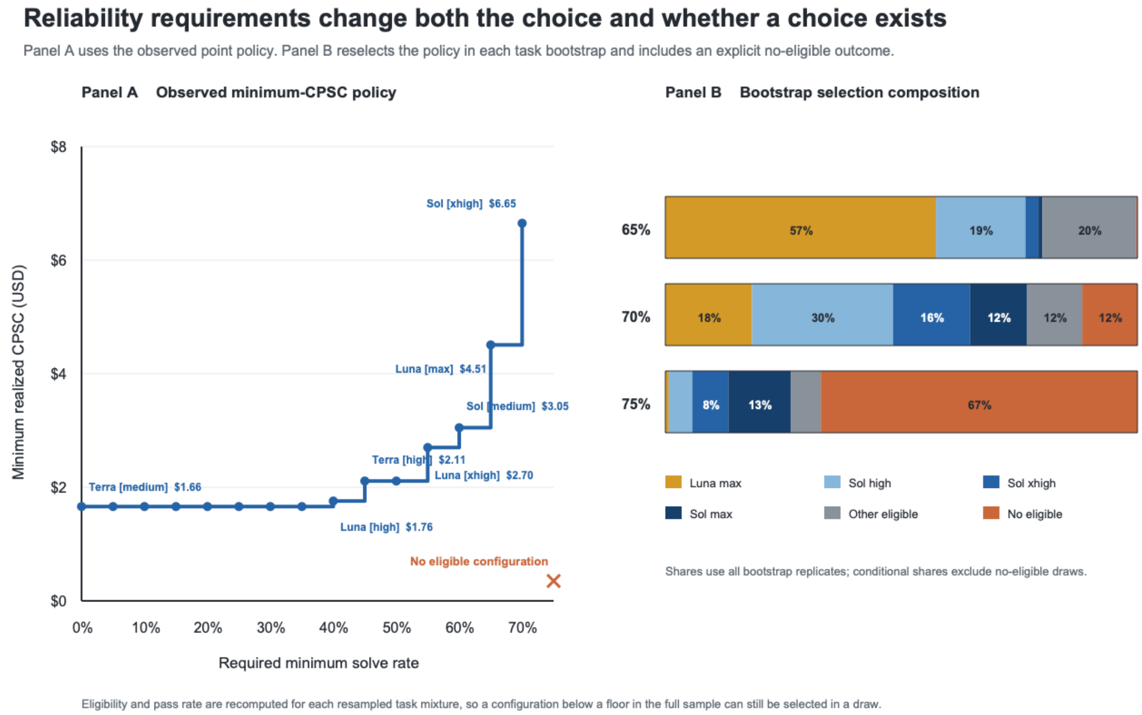

Figure 2: Exploratory reliability-floor analysis. Panel A shows the observed point policy. Panel B shows selection shares over all task-bootstrap replicates at high floors, including no-eligible draws. No configuration reaches a 75% point floor.

Table 4: Reliability-policy summary under task-cluster resampling. Selection shares are shown over all replicates and, in parentheses, conditional on a nonempty feasible set.

Floor	Point winner	Most frequent bootstrap selection	No eligible
0%	Terra medium	Terra medium, 53.5% (53.5%)	0.0%
65%	Luna max	Luna max, 57.3% (57.5%)	0.3%
70%	Sol xhigh	Sol high, 29.9% (33.8%)	11.7%
75%	None	Sol max, 13.3% (40.3%)	67.1%

This is the central identification result. DeepSWE supports economic rankings, but it does not identify an economical configuration that satisfies an unspecified production reliability contract. Repository clustering preserves and slightly strengthens the warning: no-eligible shares are 12.0% at the 70% floor and 70.0% at 75%. At sufficiently high floors, the benchmark may identify no feasible configuration at all.

5.2 Failure valuation changes the ordering

Recall that these are retrospective sensitivities, not observed failure costs. Successful attempts retain their observed spend, while each failed attempt is assigned 0.5×, 1×, or 2× a Sol-max-derived task-level spend proxy (section 3.5). Under the full-basket construction, this selects Sol medium at 0.5× and Luna max at 1× and 2× (table 5). The stricter Sol-solved 97-task variant moves the full-panel rank association toward capability rank as the charge grows. Its point minimum is Sol medium at 0.5× and 1×, then Sol high at 2×. Luna remains less expensive than Sol max at every tested multiplier: $4.45 versus $9.09 at 0.5×, $5.90 versus $9.83 at 1×, and $8.79 versus $11.30 at 2×.

Table 5: Minimum proxy-charge CPSC under two retrospective constructions. Multipliers apply to the Sol-derived task proxy; neither construction is a calibrated failure value.

Proxy construction	0.5×	1×	2×
Full 113-task basket	Sol medium	Luna max	Luna max
Sol-solved 97-task basket	Sol medium	Sol medium	Sol high

These two proxy constructions answer different counterfactuals and are intentionally kept separate. Neither is a calibrated reference budget. They show that charging failure can materially reorder the panel while the Luna–Sol pairwise reversal survives the tested Sol-favorable charges.

5.3 Equal-task weighting is required but nearly benign here

Observed-attempt pooling slightly overweights cells with four surviving attempts. Under the 113-task equal-weight estimand, the largest CPSC change is 2.58% for Luna max. No relative order among the 40 full-basket configurations changes, and the pass–CPSC frontier is identical. On the common 111-task basket, only Luna low and Luna max swap adjacent CPSC ranks.

One policy boundary does change. At an exact 35% floor, pooled pass rate makes Terra medium eligible at 35.11%, while equal-task pass rate is 34.96%; the equal-task winner is therefore Luna high. Every other five-point floor-grid winner agrees. The sensitivity does not overturn the paper, but it confirms that a declared workload estimator is necessary when eligibility is discontinuous.

The task-balanced bootstrap reaches the same high-floor conclusion. At 70%, 12.6% of replicates have no eligible configuration; at 75%, 67.8% do. Conditional selection frequencies move slightly, not qualitatively.

5.4 Cross-provider price comparisons remain soft

The artifact reports dollars across OpenAI, Anthropic, Vertex AI, Z.ai, and OpenRouter routes, but does not expose a dated, harmonized price sheet with consistent input, cache, output, and reasoning token semantics. A valid common-price reconstruction would need to reprice every candidate and any failure anchor before recomputing all metrics. We therefore report cross-provider invoice and resource surfaces as descriptive and reserve the strongest comparative claims for within-family and within-provider contrasts.

Missing-cost rules preserve the qualitative results. Charging all 126 infrastructure exclusions as failures, with missing exclusion spend imputed at the configuration mean, changes no CPSC rank, no attempt-cost frontier membership, and no winner on the tested five-point reliability grid. Exclusions remain uneven across providers, so the audit is reported rather than dismissed. The release artifact also does not encode model licensing, so the analysis compares named families and routes rather than making open-source versus proprietary claims.

6 Implications for Benchmark Design

The reanalysis supports transport of an operational accounting layer, but it also exposes four requirements for an economic benchmark:

1. **Declare the workload.** Equal-task, category-weighted, and observed-invoice estimands answer different questions when cells are incomplete.
2. **Declare reliability eligibility.** An unconstrained minimum can recommend a policy that fails most attempts, while a stringent floor can leave the feasible set empty.
3. **Value failure exogenously.** Observed failed spend is not the business consequence of a failed task. Any proxy charge should be named and stress-tested as such.
4. **Separate price from work.** Dollars, steps, tokens, latency, and compute are distinct surfaces. Provider prices are valid for buyer-invoice questions but not evidence of intrinsic computational efficiency.

DeepSWE's independent attempts leave one useful behavioral question open. A result-informed paired diagnostic sharpens the observed pattern without identifying its mechanism. On the 110 tasks with exactly four scored attempts for both Luna max and Sol max, Luna records one to three successes on 58 tasks (52.7%) versus Sol's 35 (31.8%). The paired task-bootstrap difference is 20.9 percentage points [8.2, 33.6]. A method-of-moments beta-binomial intraclass correlation is also lower for Luna, 0.353 versus 0.569, with a difference of -0.216 [-0.367, -0.054]. By contrast, the paired any-success coverage difference, 99 versus 94 tasks, is 4.5 points [-1.8, 10.9] and remains unresolved.

The supported result is therefore less-polarized task-level outcomes, not greater strategy diversity or a "chaotic" attempt process. The same counts can arise if Luna simply has more tasks with success probabilities near one half, even when its attempts use similar strategies. A bounded same-context repair loop is a different process from four independent attempts and could favor either behavior. Testing the mechanism requires accessible trajectories, declared sampling and stopping rules, and enough replications to separate rollout variance from task-mix variance.

The economic hypothesis worth testing in a separately designed workload is not that small or cheap models must win easy tasks. It is conditional: as calibrated task pressure rises, do initially inexpensive models consume enough additional repair work or failure consequence that a larger model becomes cheaper per reliable completion? DeepSWE does not exhibit the required repair protocol or failure valuation, so it motivates that question without answering it.

7 Limitations

Retrospective analysis. The public leaderboard and some aggregate inversions were inspected before the analysis plan froze. Reliability policies, resource surfaces, group-out strata, and equal-task sensitivity are logged result-informed amendments. The paper is an auditable exploratory reanalysis, not a blinded confirmatory study.

Four independent attempts. The design does not identify an infinite-retry process, a same-context repair loop, or a sequential stopping policy. Coverage and panel-solvedness are observed-sample summaries. The post-hoc dispersion diagnostic excludes three tasks without complete four-attempt cells for both focal configurations and varies the composition of the remaining 110 tasks. Its intraclass correlation is a descriptive moment estimate. Four attempts per task are too sparse to establish mechanisms such as strategy diversity.

Trajectory access. At the 2026-07-16 access check, the public release manifest and trial metadata advertised GPT-5.6 trajectories, but representative Luna and Sol objects returned S3/CloudFront `403 AccessDenied`. A Fable 5 trajectory using the same manifest pattern returned HTTP 200, indicating a selective artifact-publication failure rather than a general CDN outage.

The public repository contains no trajectory fallback. The exact reproduction and upstream issue are recorded in amendment A7, so this paper does not perform diagnosis-overlap, patch-similarity, or strategy-pivot analysis.

Task and model generalization. The bootstrap varies the composition of these 113 tasks. It does not cover new repositories, future model versions, provider changes, or production workloads. Highest-pass model rows and Pareto frontiers are selected on the same data used to describe them. The repository-cluster sensitivity preserves dependence among tasks from the same observed repository, but it does not establish generalization to unseen repositories.

Price and resource semantics. Reported dollars mix behavior with provider prices and cache rules. Steps are a harness-level proxy, not FLOPs. Recorded durations are observed elapsed-time totals, not throughput under a common concurrency or service condition. Token counters may not be harmonized across providers. The analysis cannot infer model size, cost-to-serve, or provider margin.

Failure value. Retrospective Sol-derived charges are sensitivity devices, not calibrated task budgets. DeepSWE does not observe the operational consequence of an unresolved task or the cost of replacement by a human or fallback model.

8 Conclusion

Applying an operational economic lens to DeepSWE reveals more than a cheaper leaderboard, but less than a universal winner. Realized CPSC itself is aggregate-recoverable. Trial-level records matter because they show that within Sol, max's additional dollar, resource, and time costs are resolved while its point solve-rate gain is not; that reported-dollar and recorded-work frontiers disagree; and that Luna's invoice advantage over Sol max largely vanishes when Sol is tuned to high effort. Repository-cluster resampling preserves the highlighted within-Sol and Luna–Sol contrasts and the high-floor feasibility warning.

The same records show why the answer is conditional. Reliability floors change the selected configuration and can empty the feasible set. Proxy failure charges move the ranking. Equal-task weighting is numerically modest here but changes one exact threshold decision. Cross-provider prices do not identify intrinsic efficiency. DeepSWE therefore yields several well-defined economic rankings, but it does not identify which ranking should govern an operational recommendation without a declared workload, reliability requirement, failure valuation, and price basis.

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A Descriptive Family Panel

Table 6: Highest-pass observed configuration per base model. Selection is retrospective and the table is descriptive. The last column sums input, cache, and output counters in millions per success; it is not a harmonized compute quantity.

Model	Provider	Effort	Pass	CPSC	Steps/success	Sum counters/success
GPT-5.6 Sol	OpenAI	max	72.7%	$11.54	84	21.2M
Claude Fable 5	Vertex AI	xhigh	69.9%	$19.19	98	20.8M
GPT-5.6 Terra	OpenAI	max	69.6%	$7.10	109	25.8M
GPT-5.6 Luna	OpenAI	max	67.2%	$4.51	151	44.9M
GPT-5.5	OpenAI	xhigh	67.0%	$10.78	122	24.5M
Claude Opus 4.8	Anthropic	max	59.0%	$22.42	203	57.7M
Claude Sonnet 5	Anthropic	max	53.8%	$49.03	499	268.6M
GPT-5.4	OpenAI	xhigh	51.8%	$10.92	136	35.0M
GLM-5.2	Z.ai	max	43.8%	$8.95	295	57.2M
Gemini 3.5 Flash	Vertex AI	medium	37.4%	$19.64	229	87.2M
Kimi K2.7 Code	OpenRouter	default	30.5%	$9.22	488	84.9M
Claude Sonnet 4.6	Anthropic	high	29.9%	$18.45	447	85.7M
Gemini 3.1 Pro Preview	Vertex AI	high	11.8%	$80.68	693	98.8M

B Reproducibility and Frozen Artifacts

The historical A1–A3 plan is preserved byte-for-byte in `configs/deepswe-cpsc-v0.1.json`, with its retrospective specification in `docs/deepswe-cpsc-preanalysis.md`. The A1–A5 plan is `configs/deepswe-cpsc-v0.2.json`; the executable A1–A7 plan is `configs/deepswe-cpsc-v0.4.json`. Dated amendment ledgers are `docs/deepswe-cpsc-amendments-v0.2.md`, `docs/deepswe-cpsc-amendments-v0.3.md`, and `docs/deepswe-cpsc-amendments-v0.4.md`. The plans record the input SHA-256 identities. The deterministic pipeline verifies the trial, task, and leaderboard hashes, reconciles official aggregates, runs the task-, paired-complete-task, and repository-cluster bootstraps, and hashes generated tables and figures in a manifest. Commands are in `paper/deepswe-cpsc/README.md`. Public source is at `https://github.com/lydakis/ShallowSWE`; commit `2724572` freezes the A1–A5 package. The A6–A7 revision requires its own immutable commit before public circulation.